

Milestone 1 | What previous research has established

Detecting potential false-normal predictions in celiac-related duodenal endoscopy images
 | Evidence checked 20 Sep 2026

Source / access	Image source	Unit	Label source	Method	Comparator	Measure / result	What remains unanswered
Hamza-Cherif et al. 2026 Abstract; full text paywalled	Clinical duodenal stills; 188 described, 89 normal / 75 coeliac / 24 doubtful; 164 used in binary training.	Image	Expert image annotation; histology linkage and patient IDs unverified.	Six backbones; hybrid frozen ConvNeXt-Tiny features + latent MixUp + RBF SVM; Otsu segmentation. Repeated stratified 5 x 3-fold CV.	Best end-to-end ResNet-18; other backbones and ablations. Doubtful class excluded.	Hybrid accuracy 87.39%, F1 87.28%; best end-to-end accuracy 85.19%. No reported false-normal triage metric in abstract.	Can a review flag catch positive-labeled images predicted normal? What happens to the 24 doubtful images and to correlated photos from one patient?
Scheppach et al. 2023 Abstract accessible	Routine endoscopic stills: 858 images / 182 VA patients and 846 / 323 normal-mucosa patients; external image set for testing.	Image; patients counted	VA versus normal mucosa per study; exact reference adjudication not verified in accessible abstract.	ResNet-18; external validation; 6 fellows and 4 experts; easy/difficult strata derived from fellows' consultations.	Human readers with and without AI consultation; easy versus difficult images.	External test: sensitivity 90%, specificity 76%, accuracy 84%; AI performance stable across difficulty strata.	Strong comparator for hard images, but did not test a normal-prediction referral rule. 'Difficult' is reader-defined, not model uncertainty.
Molder et al. 2023 Full text	White-light endoscopy: 66 images from 18 CD patients; 16 images from 16 controls; 567 selected patches.	100 x 100 pixel patch	Patient serology + distal-duodenal histology: CD Marsh 3 versus controls Marsh 0-1.	Artifact-filtered patches; CNN and three conventional ML approaches; 70/30 patch split.	Weighted kNN, boosted trees, bagged trees; no reported patient-disjoint test.	CNN table: sensitivity 99.30%, accuracy 98.24% on 170 test patches. Abstract reports 99.67% sensitivity.	Patch split may mix images/patients across train/test (inference, not proven). These patch metrics cannot be compared directly with image-level triage.
Shiha et al. 2024 Full narrative review	Published endoscopy and biopsy studies searched to Dec. 2023; no new image dataset.	Study / patient	Literature on visual markers versus VA on histology; CD confirmation uses serology and biopsy in context.	Narrative review of optical signs, biopsy sampling, enhanced imaging, AI.	White-light versus enhanced endoscopy in cited studies; no new classifier.	Reported white-light sensitivity for VA: 50-78.4%, especially limited in partial atrophy.	Visible VA signs are imperfect; normal-looking mucosa cannot rule out CD. Detecting VA also does not itself diagnose CD.
Kurz et al. 2022 Full systematic review PDF	22 medical-image studies (2016-2021); various modalities, no celiac cohort.	Study	Source studies' disease/reference labels vary.	Surveys MC dropout, ensembles, test-time augmentation, calibration and other uncertainty methods.	Across-study method comparisons; no shared celiac baseline.	No pooled celiac performance; surveys uncertainty and evaluation practice.	Uncertainty scores and human referral already exist; the contribution must be their rigorous celiac endoscopy evaluation.
Wei et al. 2019 Full author manuscript	Duodenal biopsy whole-slide scans: 1,230 slides / 1,048 patients; 212-slide independent test.	Patch to slide	Pathologist diagnoses: CD, normal, nonspecific duodenitis; test set reviewed by two GI pathologists.	ResNet-50 patch predictions aggregated for whole-slide classification.	Per-class diagnostic reference; patient-disjoint train/test, not an endoscopy-image benchmark.	Reported class accuracies 95.3% CD, 91.0% normal, 89.2% duodenitis; AUROC > 0.95 each.	Useful label-quality example, but microscopic tissue and macroscopic endoscopy answer different questions.
Jansson-Knodell et al. 2026 Abstract indexed; details not verified	Duodenal biopsy histopathology, not endoscopy stills.	Biopsy / patient (exact unit unverified)	Histopathology diagnosis (adjudication not verified).	AI model for biopsy interpretation; exact architecture unverified.	Not verified from accessible primary text.	Not verified from accessible primary text.	Relevant to diagnostic pathway; outside the endoscopy false-normal task. Do not use as a numeric comparator.
Geifman & El-Yaniv 2017 Method paper abstract	Generic CIFAR / ImageNet image benchmarks, no clinical images.	Image	Benchmark class labels.	Selective classification rejects uncertain predictions to trade coverage for lower error.	Ordinary classifier versus rejection at target coverage/risk.	Risk-coverage is an established measure; celiac result not applicable.	A plain confidence threshold is prior art; test whether any added cue improves false-normal capture at fixed review workload.

Where this project fits

Evaluate image-level flags on positive-labeled photos that a baseline classifier calls normal. Use low normal-class confidence as the probability baseline: in binary normal predictions, margin and entropy give the same ranking. Compare independent signals such as ensemble disagreement or test-time instability at a fixed referral workload. Choose thresholds on validation data; evaluate held-out images, grouped by patient if IDs exist. Report positive-label capture, alerts on negative-labeled images, risk/coverage and uncertainty intervals. Keep doubtful images separate unless a defensible reference label exists.

Interpretation gate: The 2026 article says 188 images (89 + 75 + 24), while the [public dataset listing](#) says 192. Reconcile actual files, duplicates, patient grouping and label provenance first. Without biopsy-linked diagnoses, call these positive-label/normal-prediction disagreements, not proven missed celiac diagnoses.